

Quantum Computing as a Means of Drastically Improving Efficiency of Deep Space Missions

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Introduction

Purpose

This report addresses the issues of time complexity in optimization problems and severe weaknesses in long distance communication currently hindering deep space exploration from advancing. Several key challenges stand in the way of development in areas of deep space exploration, and classical solutions have not yet been able to confront these problems with realistic, long term solutions. Optimization problems such as planetary classification, as well as long-distance communication challenges are common barriers outlined and addressed in this report. As opposed to classical methods of performing computations, quantum computing holds a major advantage as it utilizes the fundamental principles of quantum mechanics which only operate at a subatomic scale. As a result of these unique principles, quantum information units are capable of performing computations at remarkably fast speeds with minimal complexity. Naturally, this concept is of particular interest to nations across the world as the race for global supremacy ensues. Advances in space optimization pave the way for national space organizations and private space companies to gain a foothold in technological advancement and economic influence. This report will elaborate on current challenges faced using classical systems, examine deep space algorithm optimization methodologies proposed in the past, compare recent developments, and outline suggestions for future incorporation of quantum computing in solving complex optimization problems and improving deep space communication.

These same barriers can be framed more specifically as NP-hard optimization problems: autonomous trajectory optimization, planetary and spectral classification, space-to-Earth optical communications, and real-time decision-making all present obstacles that classical computational paradigms struggle to overcome within realistic timescales (Aldridge et al., 2011; Zainal et al.,

2019). Rather than relying on classical binary computing, quantum computing leverages the subatomic properties of superposition, entanglement, and quantum interference to process complex multivariable problem spaces (Nielsen & Chuang, 2010). This quantum advantage enables both computational speedups and quantum-enhanced signal restoration (Bennett et al., 1993), a pathway of particular relevance as global space agencies and private enterprises push toward ambitious deep space goals such as the Artemis program, Mars sample return missions, and outer-planet probes.

Background

Space, not governed by the same laws that for the most part hold the Earth's nations in unity, is open territory for development in all ways imaginable. Utilizing space for the expansion of science, global industries and national space programs is subject to the restraints of classical technological limitations. Deep space exploration presents a multitude of optimization problems and communication challenges that have yet to be fully resolved. Autonomous exploration is a key area that demands efficient optimization techniques. As spacecraft venture into uncharted territories, they must navigate complex environments, adapt to unexpected obstacles, and make critical decisions in real-time to maximize scientific discoveries. This involves optimizing trajectories, resource allocation, and sensor usage to ensure the efficient collection of data while conserving energy and minimizing risks. Classical methods have struggled to cope with the complexity and dynamic nature of these tasks, requiring more sophisticated approaches. Communication is another critical aspect that poses significant challenges. The vast distances between Earth and deep space probes result in signal propagation delays, leading to increased communication latency (the amount of time needed for signals to travel across long distances).

Deep space communication is also vulnerable to interference from cosmic rays, solar radiation, and other sources of noise, which further degrade the quality of the transmitted signals and can result in potential data loss. Optimized communication protocols and advanced error-correction techniques are needed for ensuring reliable and efficient transmission of sensitive data within or between governmental space agencies, commanding spacecraft remotely, and promoting timely decision-making. The classification of planets, stars, and moons is an optimization problem that involves analyzing vast amounts of data collected from onboard instruments. This data includes images, spectra, gravitational measurements, and other relevant parameters. Accurate and efficient classification is vital to understanding the composition, geological features, and potential habitability of planets. However, the large volume of data, coupled with the complexity of the classification algorithms, poses a significant computational challenge. Developing optimized algorithms capable of processing and analyzing large datasets in a timely manner is essential for effective planetary classification. Orbital calculations play a crucial role in mission planning and throughout their execution. Precise predictions of trajectories, gravitational interactions, and orbital transfers are necessary to ensure spacecraft reach their intended destinations and perform mission objectives effectively. Optimizing these calculations requires considering multiple variables, such as the positioning of nearby interstellar objects, gravitational forces, and changes in desired mission objectives which result in time-consuming recalculation. Classical methods often struggle with the computational complexity of these calculations, considering the sheer volume of data, unpredictable outcomes deep in space and the fact that the majority of these variables evolve dynamically.

These dynamics can be described more precisely in physical terms. Because probes traveling to outer planetary systems experience light-speed signal delays, real-time human control from Earth

becomes impossible, forcing spacecraft to optimize trajectories, adjust resource allocations, analyze telemetry, and execute hazard-avoidance maneuvers autonomously while minimizing power consumption (Aldridge et al., 2011). Signal propagation follows the inverse-square law, producing photon starvation over astronomical distances; transmissions are further disrupted by cosmic microwave background radiation, solar plasma, and interstellar dust, all of which degrade signal-to-noise ratios (SNR) and cause packet loss (Aboagye & Chen, 2021). The datasets used for planetary and exoplanet classification are similarly demanding, often including hyperspectral imagery, gravimetric measurements, and magnetic field data in addition to the images and spectra already described, while precise orbital calculations must additionally account for N-body gravitational interactions, solar radiation pressure, and trajectory-correction maneuvers (Zainal et al., 2019).

Principles of Quantum Theory and Their Applications in Quantum Computing

Quantum mechanics, a relatively newer theory introduced in the early 1900s, directly defies well-established principles belonging to theories of classical physics (Passon, 2022). One such principle, superposition, is a crucial concept in quantum computing. Traditional computing relies on classical properties of observable objects to store the smallest units of information in “bits”. Bits are limited to two states: 0 or 1. These states are typically represented as the absence or presence of some physical property such as an electric charge. Combinations of bits in varying sequences represent all information used and displayed electronically in classical systems. The “qubit” is essentially the quantum equivalent of a bit, and is represented by a subatomic particle (including electrons and photons). Similarly to bits, qubits can represent one state or the other and combine to form layers of information. Unlike bits, however, qubits have a unique property

due to their subatomic nature: they can also represent both states at the same time. A qubit in this case is referred to as being in a “superposition” of multiple states at once, demonstrating one important use case of this fundamental principle. Utilizing this property allows quantum computers to perform parallel computations and explore multiple solutions simultaneously, offering significant speed-ups in certain computational tasks compared to classical computers. By leveraging superposition, quantum algorithms can explore and manipulate a vast number of possibilities in parallel, leading to the potential for solving complex problems more efficiently.

Two additional quantum mechanical principles extend this same idea. A qubit’s state can be written formally as $|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$, meaning an n-qubit register can represent 2^n configurations simultaneously (Passon, 2022). Entanglement describes a non-local correlation between two or more qubits—commonly illustrated by the Bell state $|\Phi^+\rangle = (1/\sqrt{2})(|00\rangle + |11\rangle)$ —in which the state of one particle cannot be described independently of the other regardless of the distance separating them (Horodecki et al., 2009). This property, discussed further in the context of quantum teleportation below, underlies many proposed deep space communication protocols (Bennett et al., 1993). A third principle, quantum interference, allows quantum algorithms to manipulate the probability amplitudes α and β through constructive interference, which amplifies correct solutions, and destructive interference, which cancels out incorrect computational paths (Shor, 1994).

Definition of the Issue and Methodologies

Challenges in Computing Optimization Problems Using Classical Methods

In previous attempts to improve the logistics of deep space missions, researchers have employed a variety of classical methods with extremely limited success in navigating the barriers of

multivariable complexity. In the context of the optimization problems outlined previously, classical optimization algorithms like evolutionary algorithms, genetic algorithms, and simulated annealing have been used to find optimal solutions to combat the excessive need for computing power and time complexity. However, these algorithms often struggle to handle the intricacy and full extent of these problems due to their generalized nature and strict constraints (Aldridge, 2011).

Framed in computational complexity terms, many of these problems—trajectory pathfinding chief among them—fall under NP-hard or NP-complete complexity classes, meaning that as search spaces expand multi-dimensionally, classical heuristics such as Genetic Algorithms, Simulated Annealing, and Evolutionary Strategies become trapped in local minima or require exponential-time execution on the order of $O(2^n)$ (Aldridge et al., 2011). This limitation is compounded by onboard hardware constraints: radiation-hardened, space-grade microprocessors such as the RAD750 are typically limited to clock speeds below 200 MHz and modest memory capacity compared to ground-based systems, forcing onboard classical systems to rely on simplified models that raise operational risk and reduce mission scope.

Signal Degradation Over Long Distances

As for deep space communication, one of the major challenges faced when transferring information over long distances involves the exponential degradation of signal strength as the distance between sender and receiver increases. Another significant issue is the lack of sufficient computing power in classical hardware. Classical coding and modulation techniques have been used to enhance data transmission reliability, but these techniques have been limited to relatively short distances. Modulation schemes are one of the most common examples, and are used to

determine how to represent classical binary data in the form of signals sent and received. One primary example, phase-shift keying (PSK), is commonly used to encode and send binary information by manipulating specific orientations of outgoing signal waves (Zainal, 2019). While PSK is effective for communication on Earth, it encounters challenges when used for long-distance deep space communication. In deep space, signals travel over extremely long distances, encountering various obstacles along the way. As the signals propagate through space, they can experience interference and degradation. Factors such as cosmic rays, solar radiation, and interstellar dust particles can cause the signal to weaken and distort. The original phases that encode the information may get altered or become indistinguishable from noise due to these disturbances. As a result, the receiver may struggle to accurately decode the transmitted information. PSK also involves significant signal propagation delays (Zainal, 2019). The time it takes for the signal to travel from the sender to the receiver can be substantial due to the larger distances involved in deep space. This introduces challenges in maintaining synchronization between the signal and the receiver, since the signal may arrive at times and phases much later than anticipated. Although these techniques have been useful in traditional communication systems, they clearly face limitations when it comes to the long distances and delays involved in deep space missions.

Space-to-Earth communication today relies primarily on classical Radio Frequency (RF) links alongside emerging Deep Space Optical Communications (DSOC), which use Pulse-Position Modulation (PPM) to encode binary values as laser photon bursts; at interplanetary distances beyond roughly 1 AU, however, spatial dispersion causes severe photon loss regardless of which carrier is used (Aboagye & Chen, 2021). The table below summarizes how these classical

approaches compare with a quantum-enhanced communication architecture across several operational metrics.

Operational Metric	Classical RF Communication	Deep Space Optical Comm (DSOC)	Quantum-Enhanced Comm
Primary Carrier	Microwave (X/Ka-band)	Near-Infrared Laser Photons	Entangled Single Photons
Data Throughput	Low (10 kbps – 2 Mbps)	Moderate/High (10 – 267 Mbps)	Ultra-High Theoretical Capacity
Distance Decay	High ($1/R^2$ free-space loss)	High (Extreme photon dispersion)	Mitigated via Entanglement / Swapping
Security Mechanism	Mathematical Encryption (RSA/AES)	Mathematical Encryption	Physical Law Security (QKD)
Error Handling	Classical FEC (Reed-Solomon/LDPC)	Optical FEC Codes	Quantum Error Correction (QEC)

Planetary Classification

One common optimization problem involves planetary classification. A major limitation in identifying distant planets is the exponential growth of computational resources required as the size and complexity of the problem increase. The sheer number of variables and constraints involved in planetary classification tasks often results in an enormous search space that becomes impossible for classical algorithms to fully calculate within reasonable timeframes. As a result, classical methods tend to rely on simplifications, approximations, or heuristic approaches, which often leads to incomplete classifications. Classical algorithms struggle especially to effectively handle the ever-changing nature of planetary features. Planetary environments are subject to various complex interactions, such as geological processes, atmospheric conditions, and climate changes. Classical methods often fail to capture these dynamic relationships due to the sheer number of changing variables involved, and are limited in their ability to adapt to evolving

patterns over time. This issue is a result of the linear nature of classical computations. As a result, they are restricted in their capability to provide accurate and up-to-date classifications of planets and other bodies to researchers. When it comes to analyzing and classifying planetary data collected from remote sensing instruments, classical methods involving statistical analysis, pattern recognition algorithms, and machine learning have been used. These approaches, like support vector machines (SVM) and neural networks, help extract meaningful information from large datasets. However, their accuracy and efficiency depend on the quality of input features and the ability to handle complex data patterns. The increasing need for faster computations, efficient data transmission, and improved decision-making calls in deep space continues to grow as the demand for space domination increases globally.

Findings

Quantum Optimization Algorithms

In attempts to optimize some of these optimization challenges, researchers have explored the potential of quantum-inspired optimization algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) and the Quantum Annealing Algorithm (QAA). These algorithms aim to leverage the power of quantum parallelism and superposition to efficiently solve complex optimization problems encountered in autonomous exploration, planetary classification, and orbital calculations (Zhu, 2022). By forming these optimization challenges as mathematical optimization problems and mapping them onto quantum systems, these algorithms have shown promise in providing faster and more efficient solutions compared to classical methods.

Beyond QAOA and QAA, hybrid quantum-classical approaches such as the Variational Quantum Eigensolver (VQE) offer additional promise for deep space applications. VQE algorithms process complex spectral datasets collected by probes by efficiently mapping molecular structure models, which can accelerate exoplanet atmospheric characterization (Peruzzo et al., 2014). QAOA itself operates by alternating between a target problem Hamiltonian and a mixing Hamiltonian to solve combinatorial optimization problems such as spacecraft trajectory vectors and payload scheduling with polynomial speedups (Farhi et al., 2014; Zhu et al., 2022), while quantum annealers use transverse-field Ising configurations to navigate rugged cost landscapes and avoid local energy traps, allowing them to balance real-time operational variables such as power, thermal controls, downlink windows, and propellant usage simultaneously (Albash & Lidar, 2018).

Quantum Communication Protocols

In terms of enhancing deep space communication, Quantum key distribution (QKD) protocols are one of the most widely known theoretical solutions. When implemented under necessary conditions, they offer secure and near unbreakable cryptographic key distribution, which is crucial for protecting sensitive mission data during transmission. (National Security Agency). The National Security Agency suggests QDK may be able to “detect the presence of an eavesdropper, a feature not provided in standard cryptography” (National Security Agency). Arguments against the use of Quantum Key Distribution cite the protocol’s low tolerance for error. As is the case with the majority of quantum systems (not subject to communication protocols exclusively), stabilizing qubits requires careful noise reduction and is limited to the quality of a quantum setup’s hardware design (National Security Agency). Despite these

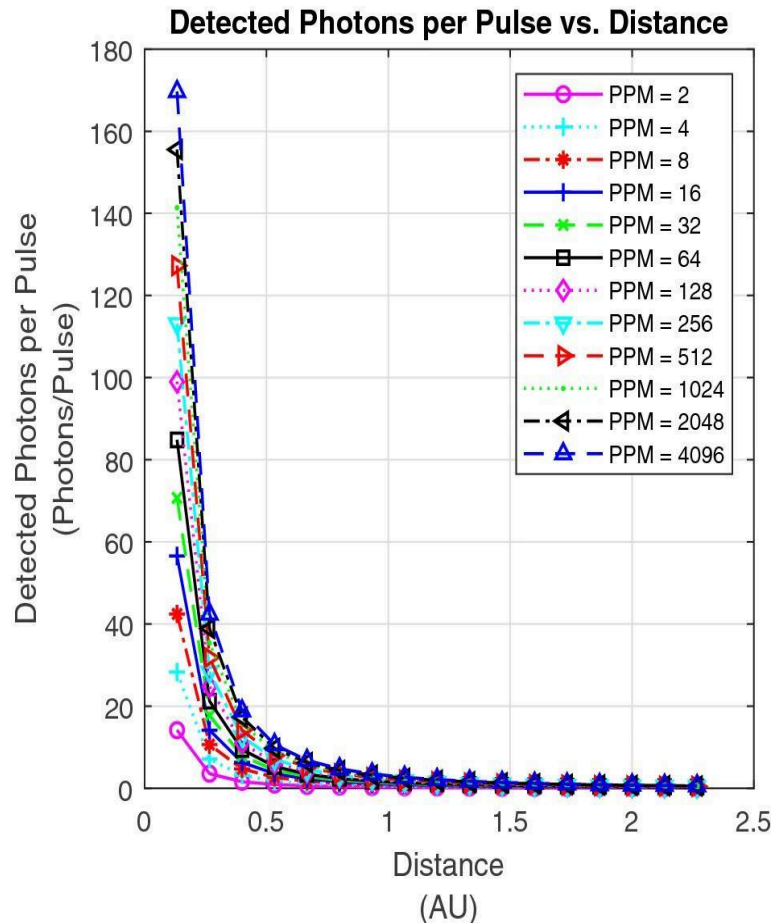
challenges, neither are impossible to solve given substantial attention to detail in the engineering and implementation processes of QKD hardware. In fact, design improvements are already beginning to take place in spite of this challenge. In 2022, researchers at the Istituto Nazionale di Ricerca Metrologica (INRIM) in Italy successfully demonstrated how single-photon sources can be utilized to significantly reduce error rates, including those presented over relatively long distances (Clivati, 2022).

Specific QKD protocols such as BB84 (Bennett & Brassard, 1984) and its decoy-state variants rely on the no-cloning theorem to establish cryptographically secure links, so that any eavesdropping attempt collapses the photon's state vector and exposes the intrusion (National Security Agency, n.d.). Satellite-to-ground demonstrations using the Micius satellite have already proven the real-world viability of satellite-based QKD (Lu et al., 2022), and twin-field QKD protocols such as the one demonstrated by Clivati et al. (2022) further show that phase stability can be maintained over long distances.

Another area of development involves the use of quantum teleportation to reduce the classical issue of signal strength degradation over long distances. Considering traditional methods of signal transfer are restricted by the laws of classical physics, signal degradation is an unresolvable issue using standard outdated techniques.

Figure 1

Detected Photons per Pulse vs. Distance Using Classic Deep Space Communication Methods



Note. The number of photons serves as a metric for measuring the decrease in signal strength over increasing distances in astronomical units. From: "Deep Space Optical Communications (DSOC) Downlink Simulation with Varying PPM Order," by Aboagye, E.D. et al., 2021, Optical and Quantum Electronics, 53(583), Figure 2, p.8. (<https://doi.org/10.1007/s11082-021-03232-z>).

The figure above demonstrates the rapid decay of information quality (through optical communication) when transferred over long distances in space. This classical method of optical information transfer does not employ quantum techniques. The use of quantum teleportation in transferring data across these distances could drastically reduce the amount of time needed for photons—information carriers in the context of optical communication—to reach target destinations during deep space missions. In the context of quantum computing, “teleportation” refers to the instantaneous retrieval of information from one quantum particle when the state of another particle is known. This is possible due to a fundamental quantum phenomenon known as

entanglement. When two quantum particles become entangled, the state of one particle cannot be described independently of the other, regardless of the distance between them. This means that measuring the properties of one particle instantly affects the properties of the other, even if they are separated by large distances, defying classical notions of locality and causality (Aboagye, 2021). By utilizing quantum entanglement, quantum teleportation allows for relatively rapid transmission of information without relying on the physical transfer of particles as is typically used. This property could be used to establish communication between Earth and deep space probes, enabling faster and more efficient data transfer without the limitations caused by signal degradation over astronomical distances. Research conducted in 2022 effectively demonstrated how quantum teleportation could be used to transmit information using entangled photon pairs. Although this concept has been explored before, these recent findings also outline an improved method of restoring data contained by photon pairs by restoring their original state, allowing for significantly lower resource consumption (Parakh, 2022). This solution essentially removes much of the uncertainty regarding cost effectiveness of communication transfer via entanglement. Teleportation protocols such as this one could potentially enable efficient and secure data transmission in deep space scenarios, instantaneously.

For the links where single photons would otherwise decay exponentially over long optical channels, Quantum Repeaters using Entanglement Swapping offer a complementary solution: rather than transmitting a payload directly, repeaters extend entanglement across long distances in stages, avoiding the photon loss that accumulates over interplanetary distances (Briegel et al., 1998).

Additionally, research efforts have been devoted to understanding quantum error correction and fault-tolerant quantum computing. As mentioned previously, deep space missions are prone to

various sources of noise (such as that from cosmic rays and radiation), which introduce errors in computations both classically and when using quantum techniques. Efficient error correction techniques are still needed for ensuring the reliability of quantum computations in the challenging deep space environment. Reed-Solomon codes, which are commonly used in classical error correction, can be understood as a special case of a quantum error correction code. These codes are already used in error-correcting algorithms for digital communication (Lie et al., 2022). Considering their applications in digital communication, there is potential for these codes to be applied to long-distance communication protocols within space.

More directly, Quantum Error Correction (QEC) addresses this problem by entangling logical qubits across multiple physical qubits, a scheme first proposed by Shor (1995). Surface Codes and Subsystem Codes built on this principle can correct the phase-flip and bit-flip errors introduced by space radiation without collapsing the underlying quantum state (Fowler et al., 2012), and recent work studying radiation effects on silicon photonics chips confirms the importance of this kind of fault tolerance for space-based QKD hardware specifically (Chen et al., 2024).

Although progress has been made in these areas, practical implementations of quantum computing for deep space algorithm optimization and communication are still in the early stages. Overcoming technical barriers, such as scalability and qubit coherence, remains a significant challenge. Despite this, ongoing research and development in quantum hardware and software continue to push the boundaries of quantum computing's potential, offering promising advantages for dramatically improving the efficiency and effectiveness of deep space missions. The solutions section of this report will expand on these issues and suggest future applications

and improvements to existing quantum approaches in both optimization problems and information transfer.

Recommendations

Up to this point, this report has examined the issues of optimization problems and communication deficits in regards to deep space exploration. Considering the wide array of partial solutions to optimization problems and communication challenges already implemented using classical computing methods, I propose the most efficient solution is to build upon existing methods using quantum optimizations. At its broadest scale, I recommend shifting time and funding away from classical systems—which, as mentioned, face limitations that cannot be resolved due to the limits of classical physics—and towards development in the most promising and well-demonstrated applications of quantum computing solutions. Quantum hardware, though generally more expensive to implement, does not necessarily need to be re-engineered and introduced into every classical system physically. Quantum algorithms, such as the Quantum Approximate Optimization Algorithm (QAOA) and the Quantum Annealing Algorithm (QAA) mentioned prior, can be accessed remotely via existing quantum computers such as those stationed at IBM. By leveraging quantum computers remotely, private space research organizations and national space agencies can build off of these algorithms to manipulate information received by existing classical hardware. Remote quantum computational operations such as these can be used to speed up optimization problems including planetary classification. In a similar manner to how these algorithms can be implemented remotely for optimization problems, quantum error correction algorithms can be employed remotely to build upon information provided by classical systems. Although the idea of quantum teleportation as a

means of instant data transfer is in its earliest stages, remote quantum error correction techniques can be used to reduce the noise prevalent in classical long-distance communications. These algorithms, albeit still in early stages, hold promise when compared to the lengthy time complexity of existing classical error correction methods. Developing quantum computers with a higher number of stabilized qubits is bound to take time, but its potential once this is achieved is not questioned. In the present, rather than replacing classical protocols completely, these solutions can be used to improve existing classical methods, allowing for optimized spending and engineering in a more feasible timescale.

These priorities can be summarized as four concrete near-term steps: (1) adopting hybrid quantum-classical architectures that pair radiation-hardened classical processors with ground-based or orbit-accessible quantum processors, offloading complex trajectory and payload optimization to cloud-hosted quantum systems via satellite links; (2) developing onboard Noisy Intermediate-Scale Quantum (NISQ) algorithms—lightweight QAOA and VQE variants capable of running on low-qubit systems for sensor data classification and autonomous navigation (Zhu et al., 2022); (3) investing in space-hardened quantum memories and single-photon sources to build orbital quantum repeaters and scalable interplanetary quantum networks (Briegel et al., 1998; Lu et al., 2022); and (4) integrating radiation-resilient, fault-tolerant surface-code QEC architectures to protect space-bound quantum hardware from cosmic radiation over the long term (Fowler et al., 2012; Chen et al., 2024).

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